

Mathematical Methods for International Commerce

Week 11/2: Lagrange Multipliers (5.6)

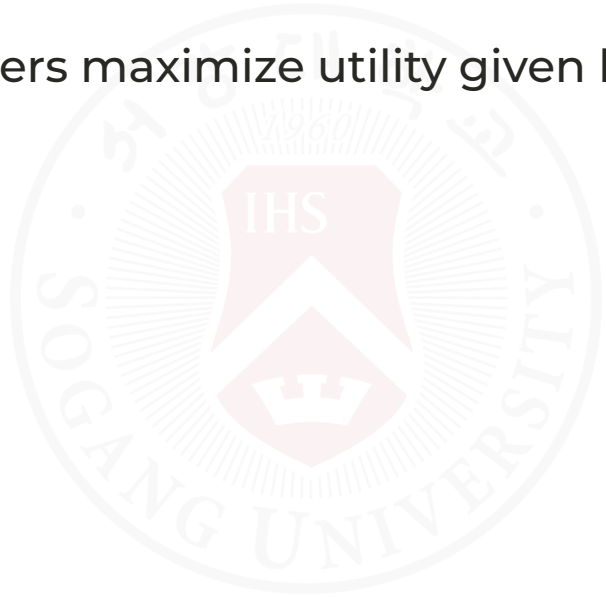
Igor Vyshnevskyi, Ph.D.

Sogang University

May 14, 2026

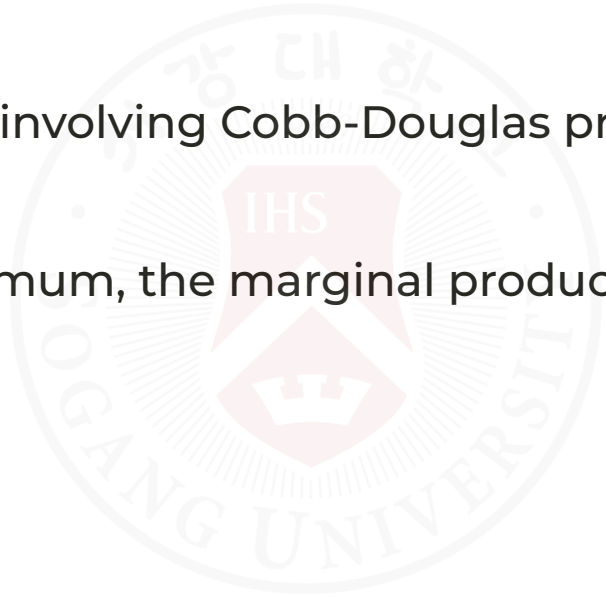
Why It Matters in Economics & Finance

- **Resource Allocation:** Optimal use of limited resources
- **Cost Minimization:** Firms aim to minimize costs given production constraints
- **Utility Maximization:** Consumers maximize utility given budget constraints



Learning Objectives

- Apply the Lagrange multiplier method to solve constrained optimization problems.
- Interpret the economic meaning of the Lagrange multiplier as the marginal value of the constraint.
- Solve optimization problems involving Cobb-Douglas production functions with cost constraints.
- Demonstrate that at the optimum, the marginal product per dollar spent is equalized across inputs.



What Are Lagrange Multipliers?

- Lagrange multipliers provide a way to find the **maximum or minimum** of a function subject to a constraint.
- They are particularly useful in economics for problems involving **cost minimization** or **utility maximization**.
- The method involves introducing a new variable (the Lagrange multiplier) to incorporate the constraint into the optimization problem.
- For a function $f(x, y)$ subject to a constraint $h(x, y) = c$, we define the Lagrangian as:

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda \cdot (c - h(x, y))$$

- The Lagrange multiplier λ can be interpreted as the **rate of change of the objective function** with respect to the constraint.

Lagrange Multipliers - Example

Problem Statement

- Optimize the objective function:

$$f(x, y) = x^2 - 3xy + 12x$$

- Subject to the constraint:

$$2x + 3y = 6$$

Step 1: Define the Lagrangian

- The Lagrangian function is given by:

$$\mathcal{L}(x, y, \lambda) = x^2 - 3xy + 12x + \lambda(6 - 2x - 3y)$$



Step 2: Derive First-Order Conditions

- Compute the partial derivatives of $\mathcal{L}$:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - 3y + 12 - 2\lambda$$

$$\frac{\partial \mathcal{L}}{\partial y} = -3x - 3\lambda$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 6 - 2x - 3y$$

Set each of these derivatives to zero to form the system of equations.

Step 3: Form the System of Equations

1. $2x - 3y - 2\lambda = -12$

2. $-3x - 3\lambda = 0$

3. $2x + 3y = 6$



Step 4: Eliminate Variables

From equation (2):

$$-3x - 3\lambda = 0 \implies x = -\lambda$$

Substitute $x = -\lambda$ into equation (3):

$$2(-\lambda) + 3y = 6 \implies -2\lambda + 3y = 6 \quad (4)$$

Substitute $x = -\lambda$ into equation (1):

$$2(-\lambda) - 3y - 2\lambda = -12 \implies -4\lambda - 3y = -12 \quad (5)$$

Now solve equations (4) and (5).

Step 5: Substitute to Find y and x

- From equation (5), substitute $\lambda = 1$:

$$-4(1) - 3y = -12$$

$$-3y = -8 \implies y = \frac{8}{3}$$

- Substitute $y = \frac{8}{3}$ and $\lambda = 1$ into equation (1):

$$2x - 3\left(\frac{8}{3}\right) - 2(1) = -12$$

$$2x - 8 - 2 = -12 \implies x = -1$$

Step 5_1: Check the second order derivative (Optional for now)

Note: We don't use Bordered Hessian here because we have only one constraint and the second derivative test is sufficient to classify the stationary point.

To classify the stationary point, use the constraint:

$$2x + 3y = 6$$

Solve for y :

$$3y = 6 - 2x$$

$$y = 2 - \frac{2}{3}x$$

Substitute this into the objective function:

$$f(x, y) = x^2 - 3xy + 12x$$

$$f(x) = x^2 - 3x \left(2 - \frac{2}{3}x \right) + 12x$$

Step 5_1: Check the second order derivative (Optional for now)

Expand:

$$f(x) = x^2 - 6x + 2x^2 + 12x$$

$$f(x) = 3x^2 + 6x$$

Now take the second derivative:

$$f'(x) = 6x + 6$$

$$f''(x) = 6$$

Since:

$$f''(x) > 0$$

the stationary point is a **constrained minimum**.

Therefore:

$$x = -1, \quad y = \frac{8}{3}$$

gives a **minimum value** of the objective function subject to the constraint.

Step 6: Calculate Optimal/Minimal Value

- Optimal solution:

- $x = -1, y = \frac{8}{3}, \lambda = 1$

- Calculate the objective function at the optimal point:

$$\begin{aligned} f\left(-1, \frac{8}{3}\right) &= (-1)^2 - 3(-1)\left(\frac{8}{3}\right) + 12(-1) \\ &= 1 + 8 - 12 = -3 \end{aligned}$$

- Stationary value of the objective function under the constraint : **-3**

Step 7: Economic Interpretation

- The Lagrange multiplier λ represents the **marginal value of the constraint**.
- In this case, it indicates how much the objective function would change if the constraint were relaxed by one unit.
- A positive λ means relaxing the constraint increases the objective value.
- A negative λ means relaxing the constraint decreases the objective value.
- The optimal solution (x, y) represents the best allocation of resources given the constraint.
- The Lagrange multiplier λ provides insight into the trade-offs involved in the optimization problem, helping to understand the sensitivity of the objective function to changes in the constraint.
- In this case, the optimal solution suggests that the firm should allocate resources in a way that maximizes the objective function while adhering to the constraint.

Lagrange Multipliers: Profit Maximization

A monopolistic producer of two goods G_1 and G_2 has the following cost function:

$$TC = 10Q_1 + Q_1Q_2 + 10Q_2$$

Demand equations:

$$P_1 = 50 - Q_1 + Q_2$$

$$P_2 = 30 + 2Q_1 - Q_2$$

Objective: Maximize profit given the constraint that the total output is 15 units.

Step 1: Formulate the Lagrangian

Objective function (Profit):

$$\pi = TR - TC$$

Total Revenue:

$$TR_1 = (50 - Q_1 + Q_2)Q_1 = 50Q_1 - Q_1^2 + Q_1Q_2$$

$$TR_2 = (30 + 2Q_1 - Q_2)Q_2 = 30Q_2 + 2Q_1Q_2 - Q_2^2$$

Total Revenue:

$$TR = 50Q_1 - Q_1^2 + 3Q_1Q_2 + 30Q_2 - Q_2^2$$

Profit function:

$$\pi = 50Q_1 - Q_1^2 + 3Q_1Q_2 + 30Q_2 - Q_2^2 - (10Q_1 + Q_1Q_2 + 10Q_2)$$

Step 1: Formulate the Lagrangian (continued)

Simplify:

$$\pi = 40Q_1 - Q_1^2 + 2Q_1Q_2 + 20Q_2 - Q_2^2$$

Constraint/Production quota:

$$Q_1 + Q_2 = 15$$

Lagrangian:

$$L(Q_1, Q_2, \lambda) = 40Q_1 - Q_1^2 + 2Q_1Q_2 + 20Q_2 - Q_2^2 + \lambda(15 - Q_1 - Q_2)$$

Step 2: First-Order Conditions

1. Derivative with respect to Q_1 :

$$\frac{\partial L}{\partial Q_1} = 40 - 2Q_1 + 2Q_2 - \lambda = 0$$

1. Derivative with respect to Q_2 :

$$\frac{\partial L}{\partial Q_2} = 20 + 2Q_1 - 2Q_2 - \lambda = 0$$

1. Derivative with respect to λ :

$$15 - Q_1 - Q_2 = 0$$

Step 3: Solving the System

Rewriting the equations:

$$1. 40 - 2Q_1 + 2Q_2 - \lambda = 0$$

$$2. 20 + 2Q_1 - 2Q_2 - \lambda = 0$$

$$3. Q_1 + Q_2 = 15$$

Step 1: Isolate (λ)

From equation (1):

$$\lambda = 40 - 2Q_1 + 2Q_2$$

From equation (2):

$$\lambda = 20 + 2Q_1 - 2Q_2$$



Step 3: Solving the System (continued)

Setting them equal:

$$40 - 2Q_1 + 2Q_2 = 20 + 2Q_1 - 2Q_2$$

Combine and simplify:

$$20 = 4Q_1 - 4Q_2$$

Divide by 4:

$$5 = Q_1 - Q_2$$



Step 3: Solving the System (continued)

Now, we have two equations:

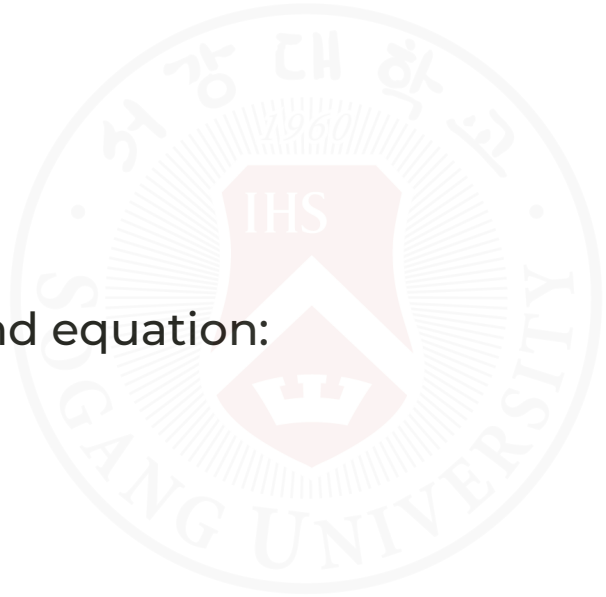
1. $Q_1 - Q_2 = 5$
2. $Q_1 + Q_2 = 15$

Adding the two equations:

$$2Q_1 = 20 \implies Q_1 = 10$$

Substitute $Q_1 = 10$ in the second equation:

$$10 + Q_2 = 15 \implies Q_2 = 5$$



Step 4: Calculate λ

Substituting $Q_1 = 10$ and $Q_2 = 5$:

$$\lambda = 40 - 2(10) + 2(5) = 30$$



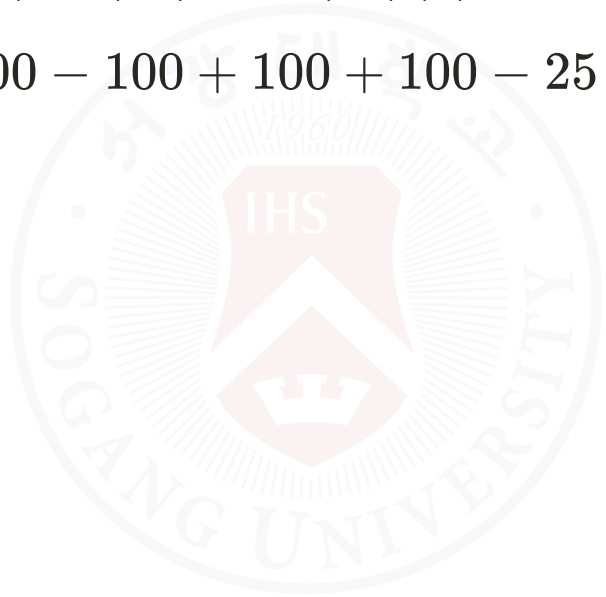
Step 5: Profit Calculation

Substitute $Q_1 = 10, Q_2 = 5$ in the profit function:

$$\pi = 40(10) - (10)^2 + 2(10)(5) + 20(5) - (5)^2$$

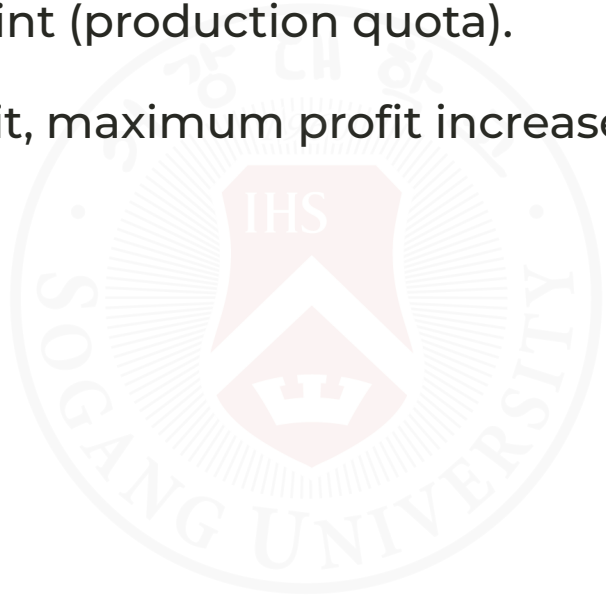
$$\pi = 400 - 100 + 100 + 100 - 25 = 475$$

Optimal Profit: 475 units.



Economic Interpretation

- The optimal profit is **475 units**.
- The Lagrange multiplier $\lambda = 30$ represents the **marginal increase in profit** for each additional unit of the constraint (production quota).
- If the quota increases by 1 unit, maximum profit increases by approximately 30 units.



Your turn

Practice Problem 1: Maximizing $2x^2 - xy$

Use Lagrange multipliers to optimise

$$-2x^2 - xy$$

subject to

$$x + y = 12$$

Give economic interpretation of the solution.

Practice Problem 2: Utility Maximization

A consumer's utility function is given by

$$U(x_1, x_2) = 2x_1x_2 + 3x_1$$

where x_1 and x_2 denote the number of items of two goods G_1 and G_2 that are bought. Each item costs \$1 for G_1 and \$2 for G_2 . Use Lagrange multipliers to find the maximum value of U if the consumer's income is \$83. Estimate the new optimal utility if the consumer's income rises by \$1.

The seal of Gangneung University is a circular emblem. It features a central shield with a crown on top and the letters 'IHS' inside. The year '1960' is inscribed above the shield. The outer ring of the seal contains the university's name in Korean and English.

2. Home work #2

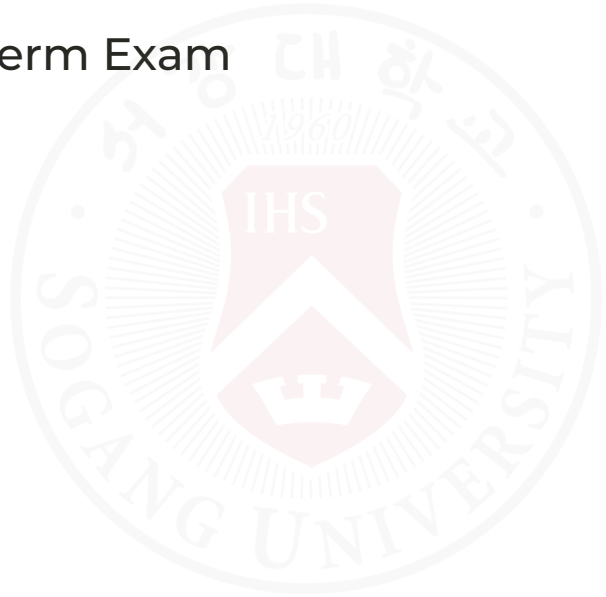
Homework #2

- **Due Date:** June 11, 2026, end of the day.
- **Submission Format:** Submit your solutions as a single PDF file via the Cyber Campus.
- **Instructions:**
 - Clearly show all steps and calculations.
 - Include explanations for your answers where applicable.
 - Ensure your submission is neat and well-organized.
 - Bring any questions to the office hours or email me.
- Work on your Home Assignment #2 (Jacques, 10th edition, Chapters 5-9):
 - Chapter 5.1: Exercise 5.1, Problem 5 (p. 411)
 - Chapter 5.2: Exercise 5.2, Problem 7 (p. 427)
 - Chapter 5.3: Exercise 5.3, Practice Problem inside the chapter (p. 432 only)
 - Chapter 5.4: Exercise 5.4, Problem 5 (p. 456)
 - Chapter 5.5: Exercise 5.5, Problems 6 and 7 (p. 467)
 - Chapter 5.6: Exercise 5.6, Problem 4 (p. 479)
 - Chapter 6.1: Exercise 6.1, Problem 3 (p. 506)
 - Chapter 6.2: Exercise 6.2, Problem 6 (p. 521)
 - Chapter 7.1: Exercise 7.1, Problem 5 (p. 553)
 - Chapter 7.2: Exercise 7.2, Problem 6 (p. 572)
 - Chapter 7.3: Exercise 7.3, Problem 5 (p. 584)
 - Chapter 8.1: Exercise 8.1, Problem 6 (p. 615)
 - Chapter 8.2: Exercise 8.2, Problem 3 (p. 626)
 - Chapter 9.1: Exercise 9.1, Problem 4 (p. 655)
 - Chapter 9.2: Exercise 9.2, Problem 3 (p. 670)

Good luck!

Next Classes

- (May 19) Indefinite Integration (6.1)
- (May 26) Quiz #2
 - Covers materials after Midterm Exam





Any QUESTIONS?

Thank you for your attention and participation!